

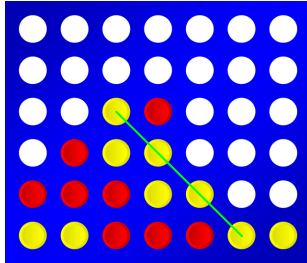
CS Topics: Mobile Application Development

Unit 1 Project: Connect 4

ALL ITEMS ARE SUBJECT TO CHANGE



Your first major project will be recreating Connect 4, the classic tabletop game from the 1980s.



Before you begin, familiarize yourself with how the game works and basic strategy by watching the following short commercials:

- [Classic Connect Four Commercial](#)
- [How to play Connect 4](#)

Game Overview: Connect 4 is a two-player game where players can take turns dropping checkers into a 6 X 7 vertical grid. Each checker falls to the lowest available space in its column and stacks on top of any previously placed checkers. The game ends when:

- One player connects **four checkers in a row** (horizontally, vertically, or diagonally)
- The board is filled

Project Overview: This project will be built using basic views. (i.e., **Button** and **TextView**). **All parts will be completed in class ONLY unless otherwise noted.**

- The **dropper** will be represented by a **TextView** that indicates where the next checker will be placed. There will be a **TextView** above each column, whose visibility we will eventually change to indicate where to drop.
- Three buttons at the bottom will control the dropper's position and when to drop it into the grid.
- You can simulate movement by showing and hiding the top row of **TextViews** as the player moves left or right.

The project will be divided into four parts.

1. **Part 1 [In Class ONLY]: Create the User Interface (UI) (2 Days, 15 points)**
 - a. Buttons to control the dropper's position & when to drop it.
 - b. A visual representation of the grid using **Buttons**.
 - c. A row of **TextViews** on the top row to indicate which column to drop into.
 - d. Labels or indicators for each player's turn.
 - e. Add a background color.
 - f. Add a font (You may use AI for this portion ONLY)

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Here is the detailed rubric for this section:

Part 1: Creating UI Rubric	
Task	PTS
The layout includes all required components (Buttons and TextViews) laid out in a grid formation.	2
XML design is not overly complex and efficiently uses layouts to create the UI.	2
Grid is visually aligned and easy to read (rows/columns are consistent).	3
Buttons are placed at the bottom to control the “dropper.”	2
TextView “dropper” widgets are present.	3
Overall design is clear and visually appealing (color, fonts, labels, margins, spacing, alignment).	3
TOTAL	15

Sample Design: Below is a sample layout to help you visualize your interface:

- Buttons at the bottom change the visibility of the “Here?” TextView to indicate where a checker will drop.
- The grid is made of TextViews (top row) and Buttons (all other rows).
Note: The row of TextViews is not in the given example. Its placement was denoted by a single TextView.
- When a checker is placed, it should occupy the lowest available spot in the selected column.
- Consider adding your own theme or style to make the project uniquely yours!
- Add a button (not seen in the example) to drop the placer into the designated position. Perhaps between the ‘move left/move right’ buttons.

